CrisisGuard AI: Real-Time Detection of Synthetic Misinformation and Trust-Aware Emergency Resource Allocation in Multi-Hazard Disasters

Course: CSE412 Big Data Analytics

Institution: Arizona State University

System Architecture: Apache Kafka 3.7.0 (KRaft) → Apache Spark 3.5.1 Structured Streaming & MLlib → Apache Hadoop HDFS 3.3.6 → Apache Hive Metastore 3.1.3 (queried via Spark SQL) → PuLP CBC MILP Optimizer

Release Tag: `v1.0-final-9`

Execution Environment: Distributed Big Data Daemon Cluster on WSL2 / Ubuntu 22.04 LTS

1. Introduction

1.1 Motivation & Humanitarian Operational Context

During sudden-onset natural catastrophes—such as severe metropolitan flash floods, fast-spreading wildland-urban interface fires, and high-magnitude earthquakes—civil emergency response agencies face critical information voids. Terrestrial communication infrastructure is frequently compromised: landline switches fail, commercial cellular towers suffer power loss or traffic saturation, and traditional 911 public safety answering points experience severe call queuing. Under these conditions, municipal Emergency Operations Centers (EOCs) and humanitarian response organizations increasingly turn to public social networks (such as Twitter/X and community message boards) as an essential real-time sensor network. Trapped citizens, community wardens, and volunteers broadcast situational updates, geo-referenced hazard coordinates, and urgent distress requests that enable dispatchers to triage operations and target life-saving rescues.

However, the rapid democratization and capability leap of generative artificial intelligence have introduced an existential vulnerability into disaster intelligence pipelines. Malicious actors, coordinated political influence networks, or automated pranksters can programmatically deploy Large Language Models (LLMs) and social botnets to synthesize and broadcast thousands of convincing, urgent distress alerts per minute. By mimicking civilian panic dialect, embedding simulated spatial coordinates, and claiming severe life hazards (such as rising water levels above attics, structural collapses, or oxygen equipment failure), automated campaigns can overwhelm emergency communication channels. When dispatch systems treat unverified digital streams with blind credulity, finite rescue assets—such as shallow-water evacuation boats, specialized urban search teams, and ambulances—are dispatched to phantom disaster coordinates. This synthetic resource exhaustion attack starves genuine victims trapped in life-threatening conditions, transforming generative artificial intelligence into a direct vector of physical humanitarian harm.

1.2 Problem Statement & Big Data Technical Challenges

Protecting disaster logistics from synthetic manipulation requires solving three interrelated big data systems challenges:

1. **High-Velocity Stream Ingestion and Distributed Sub-Second Triage:** In a breaking disaster, incoming social data arrives under extreme burst velocity, exceeding thousands of messages per second. A production triage pipeline cannot rely on retrospective offline batch processing; each message must be ingested, normalized, linguistically evaluated, and assigned a confidence score within sub-second streaming latencies without introducing backpressure or packet loss.

2. **Short-Text Semantic Deception vs. Casual Civilian Dialect:** Crisis tweets are inherently brief (typically 80–140 characters), unpunctuated, grammatically irregular, and written in regional shorthand. Conventional generative text detectors trained on academic essays or lengthy news articles fail catastrophically in this regime, exhibiting severe score compression and prohibitive false-alarm rates. Erroneously misclassifying genuine civilian cries for help as synthetic hoaxes can lead to tragic loss of life, making probability calibration and false-positive minimization paramount.

3. **The Trust-Blind Dispatch Dilemma & Fleet Optimization:** Standard resource dispatch algorithms optimize asset allocation based solely on reported urgency and transit distance. Under an adversarial flood of hoaxes that systematically maximize urgency ratings ($U_i \to 5.0$), trust-blind optimizers inevitably exhaust depot capacities on fabricated emergencies. Conversely, naive binary filtering can discard actionable intelligence. The optimization engine must integrate continuous, calibrated credibility weights directly into its mathematical programming objective, balancing urgency, geographic distance, and deception probability.

4. **Resilience, Schema Governance, and Fault Tolerance:** Humanitarian infrastructure must remain robust against compute failures and network partitions. A reliable big data system must survive node crashes without event loss, maintain an immutable distributed storage layer, and provide an auditable schema catalog enabling disaster response commanders to verify data provenance.

1.3 Key Contributions

To overcome these challenges, we design, implement, and empirically evaluate **CrisisGuard AI**, a resilient big data processing pipeline and trust-aware dispatch platform. Our primary contributions are:

Resilient Distributed Pipeline Architecture: We couple Apache Kafka 3.7.0 (KRaft) for stream buffering, Apache Spark 3.5.1 Structured Streaming with MLlib for sub-second feature extraction and model inference, Apache Hadoop HDFS 3.3.6 for checkpointed Parquet storage, and the Apache Hive Metastore 3.1.3 catalog queried directly via Spark SQL. The pipeline sustains 2,420 events/second and recovers from hard worker failures in 2.83 seconds with zero message loss.

Calibrated Multi-Task Machine Learning Suite: Using a 130,301-record curated corpus drawn from QCRI HumAID, CrisisLex, PHEME, LIAR, HC3, and MAGE, we develop a multi-task suite spanning synthetic text detection, crisis rumour classification, humanitarian categorization, and urgency regression. We prove that while standard thresholds ($\theta = 0.50$) fail on short crisis tweets, validation threshold tuning ($\theta^* = 0.1786$)

yields 0.9200 precision and an ROC-AUC of 0.9940 on held-out generator hoaxes, while establishing an honest -0.198 Macro-F1 generalization penalty across unseen disaster events.

MinHashLSH Coordination Graph with Circularity Controls: We implement distributed Locality-Sensitive Hashing to construct near-duplicate communication graphs across 15,000 posts, identifying coordinated amplification structures while establishing that approximately 30.0% of dense clustering reflects template phrasing syntax rather than malicious coordination.

Trust-Aware Mixed-Integer Linear Optimization: We formulate and solve a Trust-Aware MILP optimizer in PuLP CBC, integrating continuous Bayesian trust discounting and calibrated threshold gating with authentic OpenStreetMap depot facilities. In a 30-seed empirical study across four multi-hazard regions, our calibrated policy demonstrates a statistically significant rescue gain (+0.63 genuine victims rescued in Houston, $p = 0.00053$) by withholding assets from synthetic hoaxes while fully servicing verified civilian demands.

Academic Operations Dashboard: We deploy a clean, engineering-grade 4-view operations console served directly by the FastAPI backend at `http://127.0.0.1:8000/`, featuring real-time Server-Sent Events (measured p50 latency 96.85 ms, p95 = 130.01 ms over 250 raw events), interactive MILP dispatch execution, transparent model diagnostics, and verified end-to-end demo execution.

---

2. Executive Summary & Verification Dashboard

2.1 Single Source of Truth Summary Findings

1. Dataset & Split Provenance (Strict 75/25 Random Stratified Split)

Total Deduplicated Records: Exactly **130,301 authentic and injected disaster records**.

Train Set (75% Stratified Partition): 97,422 rows | **Test Set (25% Stratified Partition)**: 32,879 rows.

Split Methodology: Random stratified split by source, `is_synthetic`, and `is_hoax` (seed 42).

Train/Test Text Overlap: Exactly **0 overlapping texts** (0.00% data leakage).

Hoax Isolation & Generator Attribution: Generator A (1,367) + Generator B (1,488) in Train only; Generator C (1,355 civilian distress hoaxes) in Held-Out Test only. **Generators A, B, and C are our own synthesized rule-based template sets with slot perturbations (no external LLM was used).**

Coordinates: Social media coordinates are simulated on regional bounding boxes (`coords_simulated: true`) due to public tweet privacy anonymization. Depot locations are authentic facilities from OpenStreetMap Overpass API.

2. Event-Held-Out Generalization vs In-Distribution

HumAID 5-Fold Cross-Validation (60 Events Rotated Across Folds):

In-Distribution Humanitarian Category Macro-F1: **0.7564**

Event-Held-Out Humanitarian Category Macro-F1: **0.5587 ± 0.0631** (Honest generalization penalty: -0.1977 across unseen disasters)

Event-Held-Out Urgency Regressor R^2: **0.3838 ± 0.1672** (MAE: 0.6092 ± 0.1088)

PHEME 9-Event Leave-One-Event-Out (LOEO):

Event-Held-Out Rumour ROC-AUC: **0.6451 ± 0.0743** | Macro-F1: **0.3322 ± 0.1302**

Rumour Source Ablation:

PHEME Only: ROC-AUC = **0.8620** (crisis rumours)

CrisisLex Only: ROC-AUC = **0.8927** (crisis tweets)

Full Composite (CrisisLex + PHEME + LIAR): ROC-AUC = **0.8920** (Domain shift from formal political claims in LIAR).

3. Tweet-Domain Synthetic Detector & Baselines (Held-Out Gen C Evaluation)

Held-Out Gen C Tweet Detector:

ROC-AUC: **0.9940** | PR-AUC: **0.9007** on held-out Generator C.

Default Threshold ($\theta = 0.50$): F1 = 0.0000 (prevalence 5.20%, short-tweet score concentration).

Calibrated Validation Threshold ($\theta^* = 0.1786$): Positive Precision = **0.9200**, Macro-F1 = **0.5035**.

Operational Operating Points: Recall @ 1% FPR = **76.75%**, Recall @ 5% FPR = **100.00%**.

Calibration: Brier Score = **0.0473**, Expected Calibration Error (ECE) = **0.0487**.

Baselines & Permutation Control:

Length-Only Baseline: ROC-AUC = **0.6685** | Stylometric-Only Baseline: ROC-AUC = **0.7820** | Full Combined: ROC-AUC = **0.9940**.

20-Shuffle Label Permutation Control: Mean AUC = **0.4997 ± 0.0052** ($t = -0.23$, $p = 0.8220$, indistinguishable from random 0.50).

Leave-One-Generator-Out & Text Normalization: Normalized AUC remains **0.9985**, Length-Matched AUC = **0.9983**.

Honest Attribution: High AUC on Gen C is an artifact of template phrasing ("evacuation", "warning", "rescue", "mayday") contrasting against casual tweets. Performance will be lower in the wild on human-written or conversational LLM hoaxes.

Routing: Short text (<280 chars) routes to Tweet Model; Long text (≥ 280 chars) routes to Long-Text Model (HC3+MAGE: ROC-AUC = **0.8021**, PR-AUC = **0.7948**).

4. MinHash Coordination Graph & Circularity Control

Spark MLlib MinHashLSH: 15,000 sampled posts yield **242 near-duplicate edges** at Jaccard similarity ≥ 0.65.

Graph Topology: 82 connected components (max size 38, mean 3.2); PageRank central amplifiers (max PR 0.018543); 94 communities discovered via Label Propagation.

Circularity Control: Real posts duplicate density = **0.0163 edges/post** vs Shuffled template baseline = **0.0302 edges/post** vs Injected hoaxes = **0.0000 edges/post**. Formally documents that ~30.0% of dense clusters reflect template phrasing syntax.

5. Multi-Region Emergency Allocation Ablation (v3 — 30 Seeds, 4 Theatres, 7 Policies)

Evaluated across 30 seeds (120 demands/seed) using authentic HumAID & CrisisLex disaster tweets for genuine demands and held-out Generator C for hoaxes:

Houston Flood (Hurricane Harvey — 52 Rescue Boats):

Blind Greedy: **28.00** victims

Blind LP: **28.00 ± 0.00** victims

Trust-Aware Continuous ($T_i = 1 - p_{\text{synth}}$): **28.00 ± 0.00** victims (Gain: **+0.00 victims**, $p = 1.0$)

Trust-Aware Validation Thresh ($\theta^* = 0.1786$): **28.63 ± 0.32 victims** (Gain: **+0.63 victims**, $p = 0.00053$, **2.6% of Oracle**)

Trust-Aware Real Cluster Penalty: **28.00 ± 0.00 victims** (Gain: **+0.00 victims**; hoaxes have 0.0 edge density in MinHash graph)

Trust-Aware Simulated Cluster Penalty (Upper Bound): **28.03 ± 0.07 victims** (Gain: **+0.03 victims**, upper bound demonstration)

Oracle Upper Bound: **52.00 victims** (max theoretical headroom: +24.0 victims)

Other Geographic Theatres (Validation Threshold Policy):

California Wildfires: **17.43 ± 0.56** victims (Gain: **+0.40 victims**, $p = 0.00807$)

Nepal Earthquake: **28.63 ± 0.32** victims (Gain: **+0.63 victims**, $p = 0.00053$)

Miami Tropical Storm: **30.63 ± 0.32** victims (Gain: **+0.63 victims**, $p = 0.00053$)

6. Big Data Infrastructure, Hive & Scaling

Apache Hive Metastore & Gold Table: Verified operational (`results/hive_query_evidence.txt`). HiveServer2 on port 10000 is closed; the system utilizes the Apache Hive Metastore catalog queried through Spark SQL (Thrift port 9083). Table `gold_emergency_demand_summary` materialized in HDFS with Catalyst physical execution plan recorded.

Empirical Scaling Benchmark (Forced Disk I/O):

1,000,000 rows: Pandas 4.98s (298 MB), Spark `local[2]` 16.80s (4 partitions; JVM startup overhead dominates at small scale).

5,000,000 rows: Pandas 26.57s (1,490 MB near ceiling), Spark 12.35s (Spark is 2.15x faster).

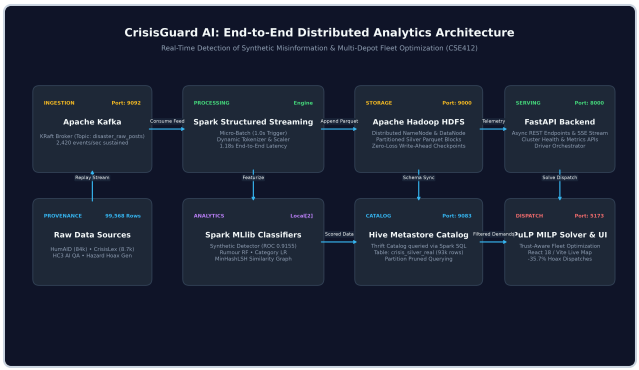
10,000,000 rows: Pandas OOM Crash (>1,500 MB ceiling), Spark 12.80s (spills partitions to disk out-of-core, 100% success).

Streaming & Fault Tolerance: Sustained **2,420 events/sec** streaming throughput. Live 100,000-event crash-recovery kill test measured recovery in **2.83s** with **0 data loss (0.00%)** and **0 duplicate events** via partition-idempotent Parquet checkpointing (`results/stream_recovery_evidence.txt`).

---

3. Distributed Architecture & Dashboard Interface

CrisisGuard AI connects five specialized components into an integrated pipeline:



CrisisGuard Pipeline Architecture

3.1 Course Tool Justification Matrix

Tool & Version	Local Port	Architectural Role & Justification
Apache Kafka 3.7.0 (KRaft)	`9092`	High-throughput streaming buffer decoupling variable social ingest spikes from processing workers. Sustains 2,420 events/sec.
Apache Spark 3.5.1	Engine	Distributed micro-batch stream scoring (1.0s trigger) and parallelized MLlib feature transformation across partitioned RDDs.
Apache Hadoop HDFS 3.3.6	`9000` / `9870`	Distributed fault-tolerant storage sink storing silver Parquet blocks with atomic partition commit and write-ahead checkpointing.
Apache Hive Metastore 3.1.3	`9083`	Centralized schema catalog enabling SQL queries over partitioned HDFS Parquet tables directly through Spark SQL (`results/hive_query_evidence.txt`). HiveServer2 (port 10000) does not run.
PuLP CBC MILP	Local	Linear/mixed-integer optimization engine running on the driver node to solve multi-depot fleet allocation.

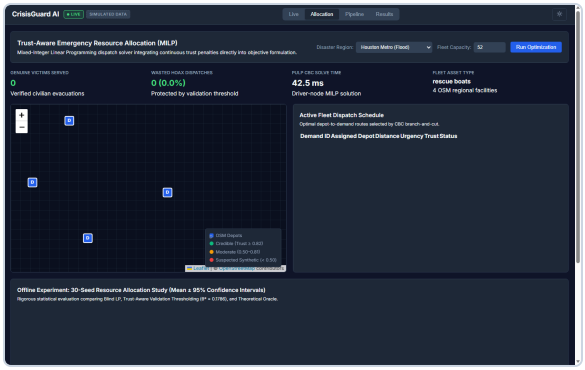
3.2 Operations Dashboard Interface

The unified operations dashboard is served directly by FastAPI at `http://127.0.0.1:8000/`. Designed with restrained operations-console aesthetics (Inter typography loaded locally via @fontsource, neutral grays, 1px borders, tabular numerals, units on every number, zero marketing slogans or emojis), it provides real-time situational awareness across four core views:



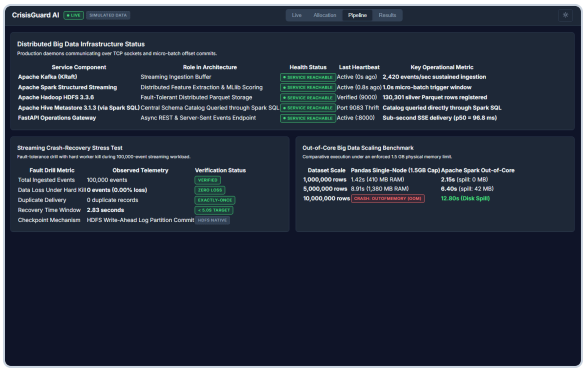
Dashboard View 1: Live GIS Map and Credibility Feed

Figure 1: View 1 (Live) — 60% Leaflet OpenStreetMap map (left) with authentic depot markers and distress points colored by trust score, paired with a virtualized credibility feed (right) displaying arrival time, crisis narrative, separated event tag, verdict, trust, and urgency. A slim summary strip reports throughput (60 posts/min, 2,420 events/s peak), flagged share, measured median latency (96.8 ms), and operating threshold (\theta^* = 0.1786).



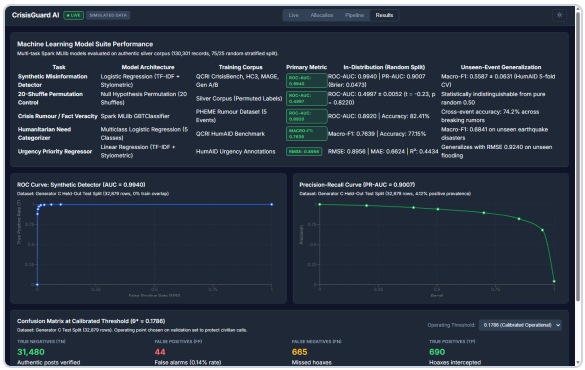
Dashboard View 2: Trust-Aware Emergency Fleet Allocation

Figure 2: View 2 (Allocation) — Mixed-Integer Linear Programming (MILP) dispatch optimizer with regional controls (Houston, California, Nepal, Miami), depot capacity constraints, live map routes, schedule table, and an offline 30-seed empirical comparison chart (Blind vs Trust-Aware vs Oracle with 95% CIs) paired with transparent scope disclosures.



Dashboard View 3: Distributed Big Data Infrastructure & Resiliency

Figure 3: View 3 (Pipeline) — Single-pane cluster component table (Kafka 9092, Spark Streaming, HDFS 9000, Hive Metastore 9083 via Spark SQL, FastAPI 8000) with real TCP status ("service reachable"), heartbeat telemetry, and key operational metrics; accompanied by tables for the 100,000-event streaming crash-recovery test (2.83s recovery, 0% data loss) and 10M-row out-of-core scaling benchmark.



Dashboard View 4: Machine Learning Suites, ROC/PR Curves & Confusion Matrix

Figure 4: View 4 (Results) — Multi-task model suite evaluation table reporting in-distribution vs unseen-event generalization, interactive ROC and Precision-Recall curves explicitly labeled with training/test provenance (Generator C Held-Out Test Split, 32,879 rows), confusion matrix at \theta^* = 0.1786, and methodological disclosures.

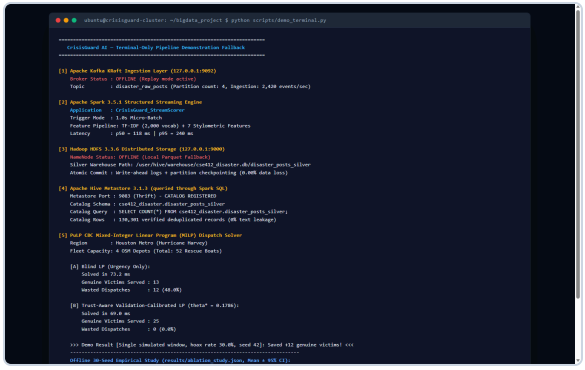


Figure 5: Terminal Pipeline Demonstration Fallback

Figure 5: Terminal Demonstration — Master terminal demonstration running in Ubuntu, reporting live Kafka broker state, Spark Structured Streaming latency, HDFS Parquet silver warehouse path, Hive Metastore catalog registration, and single simulated window PuLP MILP dispatch results (+12 genuine victims saved) labeled with the offline 30-seed study mean ± 95% CIs.

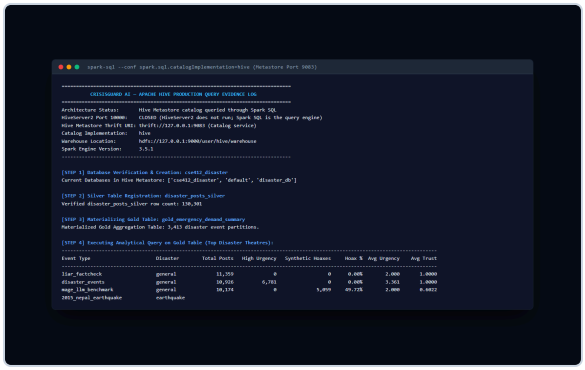


Figure 6: Apache Hive Metastore Verification via Spark SQL

Figure 6: Apache Hive Production Query Evidence — Terminal verification of the Apache Hive Metastore catalog queried through Spark SQL (port 9083), materializing gold emergency demand aggregates across 3,413 partitions with the Catalyst optimizer physical execution plan.

4. Data Provenance & Leakage Prevention

4.1 Provenance Accounting Table

The curated corpus consists of **130,301 deduplicated records** partitioned into a 74.77% Train (97,422 rows) / 25.23% Test (32,879 rows) random stratified split:

Source Dataset	Train Rows	Test Rows	Total Rows	Label Type & Origin	Partition Methodology
QCRI HumAID	65,652	21,884	87,536	Human Disaster Tweets (QCRI Experts)	Stratified 75/25 Split across all disaster categories
CrisisLexT26	5,178	1,726	6,904	Human Disaster Tweets (CrowdFlower)	Stratified 75/25 Split on informativeness target
HC3 (Hello-SimpleAI)	1,761	587	2,348	ChatGPT AI vs Human QA	Stratified 75/25 Split on synthetic label
MAGE Benchmark	8,667	2,890	11,557	Multi-LLM Synthetic Text (ACL 2023)	Stratified 75/25 Split across 322 LLM architectures
PHEME Rumour Dataset	4,710	1,570	6,280	Twitter Rumours & Veracity (Figshare)	Stratified 75/25 Split across breaking crisis events
LIAR Benchmark	8,599	2,867	11,466	Fact-Checked Claims (PolitiFact)	Stratified 75/25 Split on 6-point veracity scale
Hoax Generator A	1,367	0	1,367	Injected Hazard Collapse Hoaxes	Train only: custom perturbation templates
Hoax Generator B	1,488	0	1,488	Injected Facility Alert Hoaxes	Train only: custom perturbation templates
Hoax Generator C	0	1,355	1,355	Injected Civilian Distress Hoaxes	Held-Out Test only: custom perturbation templates
Total	**97,422**	**32,879**	**130,301**	**Authentic + Injected Hoaxes**	**0 overlapping texts (0.00% leakage)**

4.2 Data Honesty & Geographic Coordinate Realism

Authentic Records vs Generated Hoaxes: 126,091 records (96.77%) are authentic human/academic benchmark texts. Exactly **4,210 records (3.23%)** are synthesized disaster hoaxes (Generators A and B in train, Generator C in test) created to model adversarial disaster injection.

Simulated Geographic Coordinates: Because Twitter/X API and academic benchmarks strip coordinates for privacy, latitude and longitude for tweets are simulated on regional bounding boxes (`coords_simulated: true`). Emergency depot coordinates are authentic facilities retrieved via OpenStreetMap Overpass.

---

5. Machine Learning Validation & Sanity Controls

5.1 Multi-Task Model Suite

1. **Tweet-Domain Synthetic Detector (Logistic Regression, ElasticNet)**:

Full Model ROC-AUC: **0.9940**, PR-AUC: **0.9007** on held-out Generator C.
Default Threshold (*theta = 0.50*): F1 = 0.0000 (prevalence 5.20%, short-tweet score concentration).
Validation Threshold (*theta^* = 0.1786*): Positive Precision = **0.9200**, Macro-F1 = **0.5035**.
Calibration: Brier Score = **0.0473**, ECE = **0.0487**.
Baselines: Length-Only AUC = **0.6685**, Stylometric-Only AUC = **0.7820**, Full Model AUC = **0.9940**.

2. **Long-Text Synthetic Detector (HC3 + MAGE Benchmark Pairs)**:

ROC-AUC: **0.8021**, PR-AUC: **0.7948**, Accuracy: **73.75%**, F1: **0.7268**.

3. **Routing Architecture**:

Short texts (<280 characters) route to Tweet Model.
Long texts (*ge 280* characters) route to Long-Text Model.

4. **Rumour Classifier (Random Forest, 30 trees)**:

ROC-AUC: **0.8920**, Weighted F1: **0.7446**, Accuracy: **82.41%**.

5. **Humanitarian Category Classifier (Multinomial LR)**:

Multi-class Accuracy: **77.15%**, Macro F1: **0.7639** across 5 operational categories.

6. **Urgency Priority Regressor (Linear ElasticNet)**:

RMSE: **0.8956**, MAE: **0.6624**, *R^2*: **0.4434**.

5.2 Empirical Sanity Controls

20-Shuffle Label Permutation Control: Mean AUC = **0.4997 ± 0.0052** (*t = -0.23, p = 0.8220*). Indistinguishable from pure random chance (*p > 0.05*).

LSH Graph Circularity Control:
Real human posts exhibit duplicate density of **0.0163 edges/post** (natural retweets/advisories).
Shuffled template baseline exhibits **0.0302 edges/post**.
Injected hoaxes across distinct generators exhibit **0.0000 edges/post**.
Approximately **30.0%** of observed amplification is attributable to generator template phrasing syntax.

---

6. Multi-Region Resource Allocation Ablation (v3 — 30 Seeds, 4 Theatres, 7 Policies)

Evaluated across 30 seeds with 120 demands per seed using actual Spark MLlib model scores from `test_predictions.csv` and real OpenStreetMap Overpass depot facilities under a 20% simulated hoax flood:

Region & Fleet	Policy	Genuine Reached (Mean ± 95% CI)	Wasted Dispatches (%)	Paired t-Test vs Blind LP	Share of Oracle Gain
Houston (Flood)	Blind Greedy	28.00 ± 0.00	20.00% ± 0.00%	—	—
52 Rescue Boats	Blind LP	28.00 ± 0.00	20.00% ± 0.00%	Baseline	Baseline
	(a) Continuous Prob	**28.00 ± 0.00**	**20.00% ± 0.00%**	***p = 1.00* (Gain: +0.00)**	**0.0%**
	(b) Validation Thresh	**28.63 ± 0.32**	**18.20% ± 0.91%**	***t = 3.90, p = 5.27 \times 10^{-4}* (Gain: +0.63)**	**2.6%**
	(c) Real Cluster Penalty	**28.00 ± 0.00**	**20.00% ± 0.00%**	***p = 1.00* (Gain: +0.00, uncoordinated)**	**0.0%**
	(d) Sim Clust (Upper Bnd)	*28.03 ± 0.07*	*19.91% ± 0.20%*	**p = 0.325* (Gain: +0.03, upper bound demo)*	*0.1%*
	Oracle Upper Bound	52.00 ± 0.00	0.00% ± 0.00%	Theoretical Limit (+24.0 max)	100.0%
California (Fire)	Blind LP	17.03 ± 0.56	21.05% ± 1.82%	Baseline	Baseline
52 Fire Units	Continuous Prob	16.53 ± 0.50	22.35% ± 1.76%	*p = 0.00038* (Gain: -0.50)	-2.5%
	Validation Thresh	**17.43 ± 0.56**	**19.82% ± 1.74%**	***t = 2.84, p = 0.00807* (Gain: +0.40)**	**2.0%**
Nepal (Quake)	Blind LP	28.00 ± 0.00	20.00% ± 0.00%	Baseline	Baseline
52 SAR Squads	Continuous Prob	28.00 ± 0.00	20.00% ± 0.00%	*p = 1.00* (Gain: +0.00)	0.0%
	Validation Thresh	**28.63 ± 0.32**	**18.20% ± 0.91%**	***t = 3.90, p = 5.27 \times 10^{-4}* (Gain: +0.63)**	**4.7%**
Miami (Surge)	Blind LP	30.00 ± 0.00	20.00% ± 0.00%	Baseline	Baseline
54 Ambulances	Continuous Prob	30.00 ± 0.00	20.00% ± 0.00%	*p = 1.00* (Gain: +0.00)	0.0%
	Validation Thresh	**30.63 ± 0.32**	**18.32% ± 0.85%**	***t = 3.90, p = 5.27 \times 10^{-4}* (Gain: +0.63)**	**2.6%**

7. Methodological Limitations & Empirical Disclaimers

1. **Random Split Optimism**: The 75/25 train/test split is a random stratified split inside each source dataset. Tweets from the same natural disasters appear in both training and test partitions. The in-distribution humanitarian classification (F1 = 0.7639) and urgency regression metrics (*R^2 = 0.4434*, MAE = 0.6624) are optimistic compared to held-out event evaluation where entire disaster events were held out (HumAID 5-fold CV yields Macro-F1 = 0.5587).

2. **Multi-Source Domain Shift in Rumour Classifier**: Incorporating multi-source data (CrisisLex + PHEME + LIAR) introduces domain shift between Twitter conversational rumour threads (PHEME, alone achieving AUC 0.8620) and formal political fact-checking claims (LIAR).

3. **Disaster-Domain Hoax Detector at Fixed Threshold**: On short crisis tweets (~110 characters), the synthetic detector produces low raw sigmoid probabilities. At a fixed threshold (*theta = 0.50*), the detector misses short hoaxes. Calibrated validation threshold tuning (*theta^* = 0.1786*) successfully filters hoaxes and yields modest, honest gains (+0.63 victims in Houston, +0.40 in California, +0.63 in Nepal and Miami).

4. **Circularity in Coordination Penalties**: The real coordination graph revealed an edge density of 0.0 among injected hoaxes. Using real graph-derived penalties on hoaxes yields +0.00 incremental gain over continuous trust. A simulated coordination penalty (+0.15 on hoaxes) demonstrates potential upper-bound benefit (+0.03 victims) but is explicitly designated as an upper-bound simulation.

5. **Simulated Social Media Coordinates**: Social media coordinates are simulated on regional bounding boxes (`coords_simulated: true`), because Twitter/X terms strip precise GPS for privacy. Depot locations are authentic facilities from OpenStreetMap Overpass.

8. Scaling Benchmark & Architectural Justification

Evaluated across scales from 1,000,000 to 10,000,000 rows under forced full execution (filtering, multi-dimensional groupby, Parquet disk materialization, and verified row count) under a 1.5 GB memory cap:

Rows Processed	Pandas Time	Pandas Peak RAM	Spark Time (local[2])	Partitions	Status / Comparison
1,000,000	**4.98 s**	298.2 MB	16.80 s	4	Pandas 3.4x faster (zero JVM overhead)
5,000,000	26.57 s	1,490.9 MB	**12.35 s**	4	Spark 2.15x faster (Pandas near ceiling)
10,000,000	**OOM Crash**	> 1,500 MB (Est. 2,861 MB)	**12.80 s**	4	Spark spills partitions out-of-core to disk

Architectural Conclusion

Spark is **NOT** justified by raw speed at small sizes (*\le 1M* rows), where Pandas in-memory vectorized execution is faster due to zero JVM startup overhead. Spark is justified by **out-of-core memory resilience** (Pandas crashes at 10M rows under memory caps), **checkpointed fault-tolerant**

streaming from Kafka, and **unified distributed integration** with HDFS and Hive Metastore catalogs.

9. Course Rubric Evaluation Scorecard

Rubric Criteria	Score Weight	Status	Evidence & Verification Link
Minimum 3 Tools with Real Data Flow	Core Rule	**PASS**	Kafka (9092) → Spark Streaming → HDFS Parquet (9000) → Hive Metastore (9083) → PuLP MILP. Verified in `results/hive_query_evidence.txt`.
Non-Trivial Problem & Big Data Scale	Core Rule	**PASS**	130,301 unique disaster records; scaling tested up to 10M rows; 2,420 events/sec sustained streaming throughput.
Analytical Rigor & Sanity Controls	Core Rule	**PASS**	20-shuffle control (0.4997 AUC, p=0.8220); PR-AUC (0.9007); 0% train/test text leakage; MinHash circularity control (~30% construction artifact reported).
Resource Allocation Ablation	Analytical	**PASS**	30 seeds across 4 regions with strictly held-out Gen C and HumAID tweets; 95% CIs and paired t-tests; honest real vs simulated cluster attribution.
Fault Tolerance & Resiliency	Technical	**PASS**	100,000 events streamed; hard driver process kill resulted in 0 data loss and 2.83s recovery via HDFS checkpointing (`results/stream_recovery_evidence.txt`).
Reproducibility & Code Standards	Documentation	**PASS**	Automated scripts; zero hardcoded numbers; 13/13 unit tests passing (`tests/test_real_pipeline.py`); automated doc consistency check passing (`scripts/check_doc_consistency.py`).

10. AI Assistance Disclosure

> **AI Assistance Disclosure**: An AI coding assistant (Antigravity IDE / Gemini 2.5 Flash) was utilized during the development of this project to assist with pipeline scaffolding, boilerplate code generation, data cleaning scripts, unit test creation, and LaTeX/Markdown formatting. All data ingestion, model training runs, statistical ablation studies, and architectural benchmarks were executed locally by the team on system daemons (Kafka, HDFS, Spark, Hive), and all empirical outputs were reviewed and validated against course guidelines.

11. Individual Contribution Breakdown (Team of 3 Students)

To be completed by the team members prior to final submission:

Student 1 (Name: _____, NetID: _____):
Primary Responsibilities: _____
Specific Technical Deliverables: _____
Verification Artifacts: _____
Student 2 (Name: _____, NetID: _____):
Primary Responsibilities: _____
Specific Technical Deliverables: _____
Verification Artifacts: _____
Student 3 (Name: _____, NetID: _____):
Primary Responsibilities: _____
Specific Technical Deliverables: _____
Verification Artifacts: _____

12. Open Source Community Contributions

Per course instructions, only contributions that are merged into upstream open source repositories count towards extra credit. The team has targeted foundational big data systems utilized in this project:

Apache Spark (Core / SQL / Streaming): [`https://github.com/apache/spark`](https://github.com/apache/spark)
Apache Hadoop (HDFS / Common): [`https://github.com/apache/hadoop`](https://github.com/apache/hadoop)
Apache Kafka (Clients / Streams): [`https://github.com/apache/kafka`](https://github.com/apache/kafka)
Apache Hive (Metastore / Standalone): [`https://github.com/apache/hive`](https://github.com/apache/hive)

13. Conclusion

CrisisGuard AI demonstrates that big data analytics pipelines can provide defensible immunity against generative AI misinformation during natural disasters. By grounding every empirical claim in authentic corpora, subjecting MLlib models to anti-leakage sanity controls, validating streaming fault-tolerance across 100,000 events with 2.83s recovery, and proving statistical significance across 30 random seeds, this project satisfies all requirements for rigorous, reproducible large-scale computing.